



From left to right: QBO, CARE-O-BOT and KHEPERA robots

Open Source Robotics SOFTWARE PLATFORMS

Part—1

This first article in the series on open source software for robotic introduces the reader to open source robotics software platforms and focuses on the Robot Operating System (ROS).

Robotics has always lured the young and the old. Visuals of large-scale robotic cranes operating in manufacturing facilities, or futuristic movies in which robots are an integral part of human society, never fail to fire man's imagination. Robots of fantasy like Terminator, C3PO, R2D2, Wall-E, Rajnikanth's robot and Rosy the robot maid have all added to the glamour of this man-made machine.

Of late, there has been a surge in open source robotics software platforms. Of note are Player Project, Robot Operating System (ROS), MORSE (Modular Open Robots Simulation Engine), YARP (Yet Another Robot Platform), MOOS (Mission-Oriented Operating Suite), ORCOS (Open Robot Control Software) and URBI (Universal Real-time Behaviour Interface). Of all these, ROS stands out as the largest integration of such platforms, and allows the user to port in other software and elements from other platforms.

This article will introduce Linux enthusiasts to such platforms, and encourage them to try these out.

The surge in such platforms

The pioneer of the computer revolution, Bill Gates, predicted in 2006 that by the next two decades we will have a 'robot in every home'. Gates envisioned that mobile robots and

humanoid robots would become part of the urban home and help people in household chores. In the six years following that remark, some of what Gates prophesised has come true. Vacuum cleaning robots such as Roomba and Scoomba are replacing manual vacuum cleaning; automated wheelchairs for the aid of the disabled have been launched at an experimental level, and robotic assistants like PR2 and Care-O-Bot can do various domestic chores. Robotics has even entered the arena of medicine with state-of-the-art robots like the Da-Vinci and Zeus systems for tele-operated surgeries—and plenty more is being promised in the future.

Every robot is an integration of sleek electronics with versatile software. And various real-life scenarios that may be difficult to realise call for quality simulation. With so much happening in this field of technology, there is a need to set benchmarks in software for the purpose of user friendliness and versatility.

The earliest robotics software with either control or simulation interfaces was usually in two dimensions. Efforts such as Player Project and Kiks (a Matlab plug-in) unified the two paradigms; the control interface and the simulation interfaces were merged. Thus, the same code that ran on the robot would also work for the simulation. Similar efforts from